Revised dual simplex algorithm

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Pivoting operations with two systems of linear equations

Introduction: Pivoting operations with two systems of linear equations

- The dual simplex algorithm updates the **whole dictionary** at every dual pivoting operation
- The revised simplex algorithm showed that this is not necessary: two systems of linear equations give all the information needed by a pivoting operation
- In this section we do the same for the dual simplex algorithm: **which** two systems, and **what** each of them gives
- This is the version of the dual simplex algorithm used in practice, for example to re-optimize after a change of $\boldsymbol{b}$ or after adding a constraint

LP problem in standard maximization form

- We consider an LP problem in standard maximization form with $n + m$ non-negative variables and m equality linear constraints:

$$\begin{aligned} \max \quad & \boldsymbol{c}' \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} \geq 0. \end{aligned}$$

From the dual simplex to the revised dual simplex

- The dual simplex algorithm starts from a **dual feasible** basis, i.e., a basis whose dictionary satisfies $\bar{\boldsymbol{c}} \leq 0$, and it performs dual pivoting operations until the basic solution becomes feasible

- Exactly as the revised simplex algorithm does for the simplex algorithm, the **revised dual simplex algorithm**:
  - never computes the inverse matrix $\boldsymbol{A}_{\mathcal{B}}^{-1}$
  - never updates the whole dictionary

- At every iteration it determines all the information needed by a dual pivoting operation by solving **two systems of linear equations**, both of them with the same matrix $\boldsymbol{A}_{\mathcal{B}}$

1. First system of linear equations

- The dual simplex algorithm needs the values of the basic variables, in order to determine the **pivot row**
- They are the entries of the column vector $\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{b}$, which is found by solving the following system of linear equations:

$$\boldsymbol{A}_{\mathcal{B}}\,\bar{\boldsymbol{b}} = \boldsymbol{b} \quad \Longrightarrow \quad \bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{b}$$

where $\bar{\boldsymbol{b}} = (\bar{b}_i)_{i \in \mathcal{B}}$.

2. Second system of linear equations (1/2)

- Once the pivot row x_r with $r \in \mathcal{B}$ is determined, the ratio test needs the row of the matrix $\bar{\boldsymbol{A}}$ associated with x_r, that is, the values $\bar{a}_{rj}$ for $j \in \mathcal{N}$
- This row is obtained from the row vector $\boldsymbol{u} \in \mathbb{R}^{1 \times m}$, which is found by solving the following system of linear equations:

$$\boldsymbol{u}\, \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{e}'_r \quad \Longrightarrow \quad \boldsymbol{u} = \boldsymbol{e}'_r\, \boldsymbol{A}_{\mathcal{B}}^{-1}$$

2. Second system of linear equations (2/2)

- In the second system, $\boldsymbol{e}_r$ is the column vector of m entries, all equal to 0 except the one associated with the basic variable x_r, which is equal to 1
- The required row is then

$$\bar{a}_{rj} = \boldsymbol{u}\, \boldsymbol{A}_j, \qquad \forall j \in \mathcal{N}.$$

- Each entry of the row costs one scalar product between $\boldsymbol{u}$ and a column of $\boldsymbol{A}$

The same matrix, on the two sides

- The two systems of linear equations have the same matrix $\boldsymbol{A}_{\mathcal{B}}$:
  - the first one has it on the left of the unknown vector: $\boldsymbol{A}_{\mathcal{B}}\ \bar{\boldsymbol{b}} = \boldsymbol{b}$
  - the second one on its right: $\boldsymbol{u}\ \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{e}'_r$
- The vector $\boldsymbol{u}$ selects the row of $\bar{\boldsymbol{A}}$ associated with the pivot row, without computing the whole matrix $\bar{\boldsymbol{A}}$

What is still missing for the ratio test are the reduced costs $\bar{c}_j$: the next section shows how they are obtained.

Summary: Pivoting operations with two systems of linear equations

The revised dual simplex algorithm:

- Starts from a dual feasible basis ($\bar{\boldsymbol{c}} \leq 0$), as the dual simplex algorithm
- Never computes $\boldsymbol{A}_{\mathcal{B}}^{-1}$ and never updates the whole dictionary

Two systems with the same matrix $\boldsymbol{A}_{\mathcal{B}}$:

- $\boldsymbol{A}_{\mathcal{B}}\ \bar{\boldsymbol{b}} = \boldsymbol{b}$: the values of the basic variables, to choose the pivot row x_r
- $\boldsymbol{u}\ \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{e}'_r$ and $\bar{a}_{rj} = \boldsymbol{u}\ \boldsymbol{A}_j$: the row of x_r, for the ratio test

Updating the reduced costs

Introduction: Updating the reduced costs

- The ratio test of the dual simplex algorithm uses the reduced costs $\bar{c}_j$ of the non-basic variables
- We do not solve a third system to compute them: we **update** them at every iteration, with the row already given by the second system
- We also see that the update keeps the basis **dual feasible**, and we compare the two revised algorithms side by side

The reduced costs are updated, not recomputed

- The ratio test needs the reduced costs $\bar{c}_j$ with $j \in \mathcal{N}$
- They are not recomputed at every iteration: they are **updated**, exactly as the revised simplex algorithm updates the values of the basic variables
- The update uses only the row $\bar{a}_{rj}$ already determined by the second system
- At the first iteration the basis is the one of the m slack variables, and then $\boldsymbol{A}_{\mathcal{B}} = \boldsymbol{I}$ and $\boldsymbol{c}_{\mathcal{B}} = 0$, which gives

$$\bar{c}_j = c_j, \qquad \forall j \in \mathcal{N}.$$

Update of the reduced costs

After a dual pivoting operation with pivot row x_r and pivot column x_p, the reduced costs are updated as follows:

$$\bar{c}_j \longleftarrow \bar{c}_j - \frac{\bar{c}_p}{\bar{a}_{rp}} \bar{a}_{rj}, \qquad \forall j \in \mathcal{N} \setminus \{p\},$$

and

$$\bar{c}_r \longleftarrow -\frac{\bar{c}_p}{\bar{a}_{rp}},$$

where $\bar{c}_r$ is the reduced cost of the variable x_r, which leaves the basis and becomes non-basic.

The update is cheap

- The quantity

$$\frac{\bar{c}_p}{\bar{a}_{rp}}$$

  is exactly the minimum of the ratio test, which has already been computed to determine the pivot column

- The update is then a single scalar multiplication of the row $\bar{a}_{rj}$, and it costs n operations

The basis stays dual feasible

- By the choice of the pivot column,

$$\frac{\bar{c}_p}{\bar{a}_{rp}} \leq \frac{\bar{c}_j}{\bar{a}_{rj}}, \qquad \forall j \in \mathcal{N} \text{ with } \bar{a}_{rj} < 0$$

- Then the updated reduced costs remain non-positive
- The basis stays **dual feasible**, as required by the dual simplex algorithm

The two revised algorithms are symmetric

- Each of the two revised algorithms solves **two** systems of linear equations per iteration:
  - one with $\boldsymbol{A}_{\mathcal{B}}$ on the right of the unknown vector
  - one with $\boldsymbol{A}_{\mathcal{B}}$ on its left
- Each of them **updates** the other vector instead of recomputing it:
  - the revised simplex algorithm updates $\bar{\boldsymbol{b}}$
  - the revised dual simplex algorithm updates $\bar{\boldsymbol{c}}$

The next slide compares them line by line.

Revised simplex and revised dual simplex side by side

	revised simplex	**revised dual simplex**
first system	$\pi \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}}$	$\boldsymbol{A}_{\mathcal{B}} \bar{\boldsymbol{b}} = \boldsymbol{b}$
it gives	the reduced costs $\bar{\boldsymbol{c}}$	the basic values $\bar{\boldsymbol{b}}$
used to choose	the pivot column	the pivot row
second system	$\boldsymbol{A}_{\mathcal{B}} \boldsymbol{d} = \boldsymbol{A}_p$	$\boldsymbol{u} \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{e}'_r$
it gives	the column $\bar{\boldsymbol{A}}_p$	the row $\bar{a}_r$.
used for	the ratio test	the ratio test
updated, not recomputed	$\bar{\boldsymbol{b}}$	$\bar{\boldsymbol{c}}$

Summary: Updating the reduced costs

The update:

- At the first iteration $\bar{c}_j = c_j$ for every $j \in \mathcal{N}$ (basis of the slack variables)
- Then $\bar{c}_j \leftarrow \bar{c}_j - \frac{\bar{c}_p}{\bar{a}_{rp}} \bar{a}_{rj}$ for $j \in \mathcal{N} \setminus \{p\}$ and $\bar{c}_r \leftarrow -\frac{\bar{c}_p}{\bar{a}_{rp}}$
- $\frac{\bar{c}_p}{\bar{a}_{rp}}$ is the minimum of the ratio test: the update costs n operations and keeps the basis dual feasible

Symmetry:

- Both revised algorithms solve two systems with $\boldsymbol{A}_{\mathcal{B}}$ and update the other vector: $\bar{\boldsymbol{b}}$ in the revised simplex, $\bar{\boldsymbol{c}}$ in the revised dual simplex

Pseudo-code of the revised dual simplex algorithm

Introduction: Pseudo-code of the revised dual simplex algorithm

- We now put the two systems and the update of the reduced costs together in the **pseudo-code** of the algorithm
- The input is an LP problem with $\boldsymbol{c} \leq 0$: the basis of the slack variables is then dual feasible, and the algorithm can start from it
- Each iteration has a **feasibility check** (first system) and, if needed, a **dual pivoting operation** (second system, ratio test, update)

Revised dual simplex algorithm (1/4)

Input

An LP problem in standard maximization form obtained from an LP problem in canonical maximization form with $\boldsymbol{c} \leq 0$ by adding the m slack variables, so that the basis of the slack variables is **dual feasible**.

Output

An optimal solution of the LP problem, if one exists; otherwise, a certificate that the LP problem is infeasible.

Step 0: **Initialization**

Consider the basis of the m slack variables, for which $\boldsymbol{A}_{\mathcal{B}} = \boldsymbol{I}$ and $\bar{c}_j = c_j \leq 0$ for every $j \in \mathcal{N}$, i.e., the basis is dual feasible. Go to Step 1.

Revised dual simplex algorithm (2/4)

Step 1: **Feasibility check**

a. Solve $\boldsymbol{A}_{\mathcal{B}}\, \bar{\boldsymbol{b}} = \boldsymbol{b}$ to compute $\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b}$.

If $\bar{\boldsymbol{b}} \geq 0$, then the basic feasible solution

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \quad \text{and} \quad \boldsymbol{x}_{\mathcal{N}} = 0$$

is an optimal solution, of optimal value

$$\bar{z} = \boldsymbol{c}'_{\mathcal{B}}\, \bar{\boldsymbol{b}}.$$

STOP.

Else go to Step 2.

Revised dual simplex algorithm (3/4)

Step 2: **Dual pivoting operations**

The **pivot row** is the basic variable x_r with the smallest index $r \in \mathcal{B}$ among those with $\bar{b}_r < 0$.

a. Solve $\boldsymbol{u}\, \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{e}'_r$ and compute $\bar{a}_{rj} = \boldsymbol{u}\, \boldsymbol{A}_j$ for every $j \in \mathcal{N}$.

b. If $\bar{a}_{rj} \geq 0$ for every $j \in \mathcal{N}$, then the LP problem is **infeasible**. `STOP`.

c. Determine the **pivot column**, i.e., a non-basic variable x_p with $p \in \mathcal{N}$ such that

$$\frac{\bar{c}_p}{\bar{a}_{rp}} = \min \left\{ \frac{\bar{c}_j}{\bar{a}_{rj}} : j \in \mathcal{N},\ \bar{a}_{rj} < 0 \right\};$$

in case of ties, the non-basic variable with the smallest index is chosen.

Revised dual simplex algorithm (4/4)

Step 2: **Dual pivoting operations** (continued)

d. Update the reduced costs:

$$\bar{c}_j \leftarrow \bar{c}_j - \frac{\bar{c}_p}{\bar{a}_{rp}}\,\bar{a}_{rj}, \quad \forall j \in \mathcal{N} \setminus \{p\}, \qquad \text{and}$$

$$\bar{c}_r \leftarrow -\frac{\bar{c}_p}{\bar{a}_{rp}}.$$

e. Update the basis with the new partition of the indices of the variables:

$$\mathcal{B} \leftarrow \big(\mathcal{B} \setminus \{r\}\big) \cup \{p\} \quad \text{and}$$

$$\mathcal{N} \leftarrow \big(\mathcal{N} \setminus \{p\}\big) \cup \{r\}.$$

Go to Step 1.

The dictionary is never updated

- The revised dual simplex algorithm does not update the dictionary
- At every iteration it **updates** only:
  - the partition $\mathcal{B}$ and $\mathcal{N}$
  - the reduced costs $\bar{\boldsymbol{c}}$
- and it **recomputes**, by solving the two systems of linear equations:
  - the values of the basic variables $\bar{\boldsymbol{b}}$
  - the row $\bar{a}_{r\cdot}$ of the pivot row

Summary: Pseudo-code of the revised dual simplex algorithm

Start:

- $\boldsymbol{c} \leq 0$: the basis of the slack variables, with $\boldsymbol{A}_{\mathcal{B}} = \boldsymbol{I}$ and $\bar{c}_j = c_j$, is dual feasible

Every iteration:

- Step 1: solve $\boldsymbol{A}_{\mathcal{B}}\, \bar{\boldsymbol{b}} = \boldsymbol{b}$; if $\bar{\boldsymbol{b}} \geq 0$ the basis is optimal
- Step 2: pivot row x_r with the smallest index among $\bar{b}_r < 0$; solve $\boldsymbol{u}\, \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{e}'_r$; if all $\bar{a}_{rj} \geq 0$ the problem is infeasible; otherwise ratio test, update of $\bar{\boldsymbol{c}}$ and of the partition

Execution of the revised dual simplex algorithm

Introduction: Execution of the revised dual simplex algorithm

- We now run the algorithm, step by step, on an LP problem with $n = 2$ variables and $m = 3$ constraints
- At every iteration we write the matrix $\boldsymbol{A}_{\mathcal{B}}$, we solve the two systems and we update the reduced costs
- After **three** iterations we reach an optimal basis, and with one more system we also find the optimal solution of the dual problem

The LP problem of the examples

- We consider the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 3$ less-than-or-equal-to linear constraints, which is solved in the examples of this section:

$$
\begin{array}{rrrrrrrr}
\max & - & x_1 & - & 3 & x_2 & & \\
\text{subject to} & - & x_1 & - & 2 & x_2 & \leq & -6 \\
 & & x_1 & - & 4 & x_2 & \leq & -6 \\
 & 2 & x_1 & - & 2 & x_2 & \leq & 3 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$

The standard maximization form

- By adding $m = 3$ slack variables we obtain its standard maximization form, with

$$\boldsymbol{c} = \begin{pmatrix} -1 \\ -3 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix} -1 & -2 & 1 & 0 & 0 \\ 1 & -4 & 0 & 1 & 0 \\ 2 & -2 & 0 & 0 & 1 \end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix} -6 \\ -6 \\ 3 \end{pmatrix}$$

A dual feasible initial basis

- Since $\boldsymbol{c} \leq 0$, the partition

$$\mathcal{B} = \{3, \ 4, \ 5\} \qquad \text{and} \qquad \mathcal{N} = \{1, \ 2\}$$

is **dual feasible**, with reduced costs

$$\bar{c}_1 = c_1 = -1 \qquad \text{and} \qquad \bar{c}_2 = c_2 = -3.$$

- Since $\boldsymbol{b} \not\geq 0$, the same basis is not primal feasible: the revised dual simplex algorithm can start from it

Example: first iteration (1/5)

- With $\mathcal{B} = \{3, \ 4, \ 5\}$ and $\mathcal{N} = \{1, \ 2\}$ we have

$$\boldsymbol{A}_{\mathcal{B}} = \boldsymbol{I}$$

- The reduced costs are

$$\bar{c}_1 = -1 \qquad \text{and} \qquad \bar{c}_2 = -3$$

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Example: first iteration (2/5)

First system. We solve $\boldsymbol{A}_{\mathcal{B}}\ \bar{\boldsymbol{b}} = \boldsymbol{b}$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \bar{\boldsymbol{b}} = \begin{pmatrix} -6 \\ -6 \\ 3 \end{pmatrix} \qquad \Longrightarrow \qquad \bar{\boldsymbol{b}} = \begin{pmatrix} -6 \\ -6 \\ 3 \end{pmatrix}$$

- Since $\bar{b}_3 = -6 < 0$, the basis is not primal feasible
- Among x_3 and x_4, both with value -6, we choose the one with the smallest index, x_3, as **pivot row**

Example: first iteration (3/5)

Second system.

- The basic variable x_3 is the first one of $\mathcal{B}$, and then $\boldsymbol{e}_3 = (1 \;\; 0 \;\; 0)'$
- We solve $\boldsymbol{u}\, \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{e}_3'$ and obtain $\boldsymbol{u} = (1 \;\; 0 \;\; 0)$
- The row of $\bar{\boldsymbol{A}}$ associated with x_3 is then

$$\bar{a}_{31} = \boldsymbol{u}\, \boldsymbol{A}_1 = -1 \qquad \text{and} \qquad \bar{a}_{32} = \boldsymbol{u}\, \boldsymbol{A}_2 = -2,$$

both negative, and then the LP problem is not detected as infeasible

Example: first iteration (4/5)

Ratio test.

- Both non-basic variables are candidates:

$$\min \left\{ \underbrace{\frac{-1}{-1}}_{=1}, \underbrace{\frac{-3}{-2}}_{=\frac{3}{2}} \right\} = 1$$

- The **pivot column** is then x_1

Example: first iteration (5/5)

Update of the reduced costs.

- With $\frac{\bar{c}_1}{\bar{a}_{31}} = 1$ we obtain

$$\bar{c}_2 \leftarrow \bar{c}_2 - 1 \cdot \bar{a}_{32} = -3 - 1 \cdot (-2) = -1$$

and

$$\bar{c}_3 \leftarrow -1 = -1$$

- The new partition is

$$\mathcal{B} = \{1, \ 4, \ 5\} \qquad \text{and} \qquad \mathcal{N} = \{2, \ 3\}$$

Example: second iteration (1/5)

- With $\mathcal{B} = \{1, \ 4, \ 5\}$ and $\mathcal{N} = \{2, \ 3\}$ we have

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} -1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

- The reduced costs, updated at the previous iteration, are

$$\bar{c}_2 = -1 \qquad \text{and} \qquad \bar{c}_3 = -1$$

Example: second iteration (2/5)

First system. We solve $\boldsymbol{A}_{\mathcal{B}}\, \bar{\boldsymbol{b}} = \boldsymbol{b}$, that is,

$$\begin{cases} -\bar{b}_1 = -6 \\ \bar{b}_1 + \bar{b}_4 = -6 \\ 2\,\bar{b}_1 + \bar{b}_5 = 3 \end{cases} \qquad \Longrightarrow \qquad \bar{\boldsymbol{b}} = \begin{pmatrix} 6 \\ -12 \\ -9 \end{pmatrix}$$

- Two basic variables have a negative value
- We choose the one with the smallest index, x_4, as **pivot row**

Example: second iteration (3/5)

Second system.

- The basic variable x_4 is the second one of $\mathcal{B}$, and then $\boldsymbol{e}_4 = (0 \;\; 1 \;\; 0)'$
- We solve $\boldsymbol{u}\, \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{e}_4'$, that is,

$$\begin{cases} -u_1 + u_2 + 2\,u_3 = 0 \\ u_2 = 1 \\ u_3 = 0 \end{cases} \qquad \Longrightarrow \qquad \boldsymbol{u} = (1 \;\; 1 \;\; 0)$$

- The row of $\bar{\boldsymbol{A}}$ associated with x_4 is

$$\bar{a}_{42} = \boldsymbol{u}\, \boldsymbol{A}_2 = -2 - 4 = -6 \qquad \text{and} \qquad \bar{a}_{43} = \boldsymbol{u}\, \boldsymbol{A}_3 = 1$$

Example: second iteration (4/5)

Ratio test.

- The only candidate is x_2, since $\bar{a}_{42} = -6 < 0$ while $\bar{a}_{43} = 1 \geq 0$:

$$\frac{\bar{c}_2}{\bar{a}_{42}} = \frac{-1}{-6} = \frac{1}{6}$$

- The **pivot column** is then x_2

Example: second iteration (5/5)

Update of the reduced costs.

- With $\frac{\bar{c}_2}{\bar{a}_{42}} = \frac{1}{6}$ we obtain

$$\bar{c}_3 \leftarrow \bar{c}_3 - \frac{1}{6} \cdot \bar{a}_{43} = -1 - \frac{1}{6} = -\frac{7}{6}$$

and

$$\bar{c}_4 \leftarrow -\frac{1}{6}$$

- The new partition is

$$\mathcal{B} = \{1, \ 2, \ 5\} \qquad \text{and} \qquad \mathcal{N} = \{3, \ 4\}$$

Example: third iteration (1/4)

- With $\mathcal{B} = \{1, \ 2, \ 5\}$ and $\mathcal{N} = \{3, \ 4\}$ we have

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} -1 & -2 & 0 \\ 1 & -4 & 0 \\ 2 & -2 & 1 \end{pmatrix}$$

- The reduced costs are

$$\bar{c}_3 = -\frac{7}{6} \qquad \text{and} \qquad \bar{c}_4 = -\frac{1}{6}$$

Example: third iteration (2/4)

First system. We solve $\boldsymbol{A}_{\mathcal{B}}\, \bar{\boldsymbol{b}} = \boldsymbol{b}$, that is,

$$\begin{cases} -\bar{b}_1 - 2\,\bar{b}_2 = -6 \\ \bar{b}_1 - 4\,\bar{b}_2 = -6 \\ 2\,\bar{b}_1 - 2\,\bar{b}_2 + \bar{b}_5 = 3 \end{cases} \qquad \Longrightarrow \qquad \bar{\boldsymbol{b}} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$$

- Now $\bar{\boldsymbol{b}} \geq 0$: the basis is primal feasible and dual feasible, and then it is **optimal**

Example: third iteration (3/4)

- No further system has to be solved: the optimal value is the scalar product

$$\bar{z} = \boldsymbol{c}'_{\mathcal{B}}\, \bar{\boldsymbol{b}} = (-1) \cdot 2 + (-3) \cdot 2 + 0 \cdot 3 = -8$$

- An optimal solution of this LP problem is

$$\tilde{\boldsymbol{x}}(\, LP\,) = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

and the optimal value is $z(\, LP\,) = -8$

Example: third iteration (4/4)

- If the optimal solution of the **dual problem** is also required, it is obtained with one further system of linear equations, solved once at the end and not at every iteration:

$$\boldsymbol{\pi}\, \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}} \qquad \Longrightarrow \qquad \boldsymbol{\pi} = \begin{pmatrix} \frac{7}{6} & \frac{1}{6} & 0 \end{pmatrix}$$

- That is,

$$\tilde{\boldsymbol{\pi}}(\, D(LP)\,) = \begin{pmatrix} \frac{7}{6} & \frac{1}{6} & 0 \end{pmatrix}',$$

whose objective value is $\boldsymbol{b}'\tilde{\boldsymbol{\pi}} = -8 = z(\, LP\,)$, in accordance with the Strong Duality theorem

Summary: Execution of the revised dual simplex algorithm

Three iterations:

- $\mathcal{B} = \{3, \ 4, \ 5\}$: pivot row x_3, pivot column x_1
- $\mathcal{B} = \{1, \ 4, \ 5\}$: pivot row x_4, pivot column x_2
- $\mathcal{B} = \{1, \ 2, \ 5\}$: $\bar{\boldsymbol{b}} = (2 \ \ 2 \ \ 3)' \geq 0$, optimal

Result:

- $\tilde{\boldsymbol{x}}(\, LP\,) = (2 \ \ 2)'$ and $z(\, LP\,) = -8$
- One more system $\boldsymbol{\pi}\, \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}}$, solved once at the end, gives $\tilde{\boldsymbol{\pi}}(\, D(LP)\,) = \left(\frac{7}{6} \ \ \frac{1}{6} \ \ 0\right)'$, with the same value -8

Summary of the chapter (1/2)

Two systems of linear equations per iteration:

- $\boldsymbol{A}_{\mathcal{B}}\,\bar{\boldsymbol{b}} = \boldsymbol{b}$ gives the values of the basic variables and the pivot row x_r
- $\boldsymbol{u}\,\boldsymbol{A}_{\mathcal{B}} = \boldsymbol{e}'_r$ gives the row $\bar{a}_{rj} = \boldsymbol{u}\,\boldsymbol{A}_j$ for the ratio test
- No inverse matrix and no dictionary

Reduced costs:

- Updated with the minimum ratio $\frac{\bar{c}_p}{\bar{a}_{rp}}$, in n operations, keeping the basis dual feasible

Summary of the chapter (2/2)

Symmetry with the revised simplex algorithm:

- The revised simplex algorithm recomputes $\bar{\boldsymbol{c}}$ and the column $\bar{\boldsymbol{A}}_p$, and updates $\bar{\boldsymbol{b}}$
- The revised dual simplex algorithm recomputes $\bar{\boldsymbol{b}}$ and the row $\bar{a}_{r\cdot}$, and updates $\bar{\boldsymbol{c}}$

Stopping:

- $\bar{\boldsymbol{b}} \geq 0$: optimal basis, of value $\bar{z} = \boldsymbol{c}'_{\mathcal{B}}\, \bar{\boldsymbol{b}}$
- $\bar{a}_{rj} \geq 0$ for every $j \in \mathcal{N}$: the LP problem is infeasible
- The optimal dual solution costs one more system, solved once at the end